

- 8 A lichen describes a mutually beneficial association of a fungus with an organism termed a photobiont. An example of a photobiont is the green alga, *Trebouxia* sp., which is a photosynthetic protocist.

Fig. 8.1 shows lichen attached to a tree.



Fig. 8.1

- (a) Outline the characteristic features of the kingdom Fungi.

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- (b) Suggest how a fungus **and** a green alga benefit from the relationship shown by a lichen.

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- (c) A suspension of *Trebouxia* in water was used to investigate the effect of the intensity of light on the rate of photosynthesis. The volume of oxygen released over a set period of time was used as a measure of the rate of photosynthesis at each different light intensity.

All other conditions were kept constant.

Fig. 8.2 shows how light intensity affected the volume of oxygen released by *Trebouxia*.

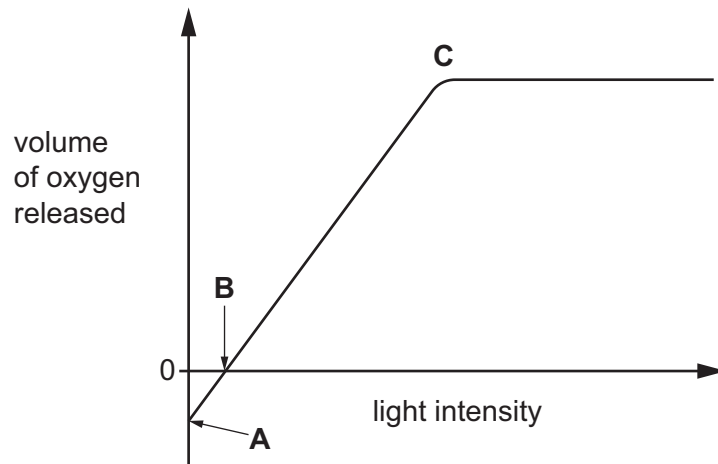


Fig. 8.2

- (i) State which photosystem is involved in the release of oxygen.

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- (ii) With reference to Fig. 8.2:

- explain the curve between **A** and **B**
- explain why the curve levels off after **C**.

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